

# DATA SCIENCE RESOURCES

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## Supervised Learning Projects

### Project 1: Credit Card Default Prediction

**Problem Statement:** Financial institutions must assess the risk of loan or credit card default to minimize losses and improve lending strategies.

**Objective:** Build a classification model to predict whether a customer is likely to default on their credit card payment.

#### Dataset Source

- [UCI Credit Card Default Dataset](#)
- [Kaggle credit risk datasets](#)

#### Techniques and Concepts

- Binary classification
- Feature engineering (repayment history, credit utilization)
- Handling class imbalance (SMOTE, undersampling)
- Logistic Regression, Random Forest, XGBoost

**Tools and Libraries:** Python, pandas, scikit-learn, imbalanced-learn, matplotlib, XGBoost

**Expected Outcome:** A model with performance metrics like AUC-ROC, precision-recall that helps banks proactively manage high-risk clients.

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## Project 2 : Diabetes Prediction Using Machine

**Problem Statement:** Early detection of diabetes can significantly reduce the impact of the disease, especially in high-risk populations.

**Objective:** Develop a model that predicts the presence of diabetes based on diagnostic health parameters.

**Dataset Source:** [Pima Indians Diabetes Dataset \(UCI\)](#)

### Techniques and Concepts

- Binary classification
- Feature scaling and outlier treatment
- Logistic Regression, SVM, Decision Trees
- Evaluation using accuracy, recall, F1-score

**Tools and Libraries:** Python, pandas, scikit-learn, seaborn, matplotlib

**Expected Outcome:** A prediction model usable by healthcare providers to flag potential diabetes cases early.

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## Project 3 : House Price Prediction (Advanced Regression)

**Problem Statement:** Home buyers and real estate businesses need accurate house price estimations to make informed buying, selling, and investment decisions.

**Objective:** Predict housing prices based on features such as location, square footage, number of rooms, and amenities.

**Dataset Source:** [House Prices: Advanced Regression Challenge](#)

## Techniques and Concepts

- Regression modeling
- Missing value treatment and one-hot encoding
- Feature selection and cross-validation
- XGBoost, Lasso Regression, Ridge Regression

Tools and Libraries: Python, pandas, numpy, scikit-learn, XGBoost, LightGBM

**Expected Outcome:** An accurate regression model with RMSE as the performance metric, usable for dynamic price estimation apps.

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## Project 4 : Email Spam Detection

**Problem Statement:** Email service providers need to identify and filter spam emails without blocking important user messages.

**Objective:** Classify emails as spam or not spam based on their content, metadata, and structure.

## Dataset Source

- [SpamAssassin Public Corpus](#)
- [Kaggle SMS Spam Collection](#)

## Techniques and Concepts

- Natural Language Processing (NLP)
- Text preprocessing (tokenization, stopword removal, TF-IDF)
- Naive Bayes, Logistic Regression, SVM

- Evaluation using confusion matrix and ROC-AUC

Tools and Libraries: Python, scikit-learn, NLTK, pandas, seaborn

Expected Outcome: A lightweight spam detection engine that can be deployed in real-time email systems.

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### **Project 5: Heart Disease Risk Classifier**

**Problem Statement:** Cardiovascular disease is a leading cause of death globally. Predictive models can save lives by enabling early diagnosis.

**Objective:** Predict the likelihood of a patient having heart disease based on diagnostic features such as age, cholesterol, and resting ECG.

**Dataset Source:** [UCI Heart Disease Dataset](#)

#### **Techniques and Concepts**

- Binary classification
- Data normalization and correlation filtering
- Logistic Regression, Random Forest, KNN
- Precision-recall and ROC curve analysis

Tools and Libraries: Python, pandas, seaborn, scikit-learn, matplotlib

Expected Outcome: A clinically useful model for screening high-risk patients using routine medical data.

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